SOCIAL INTERNSHIP

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TOPIC- URBAN FLOODING IN GUWAHATI: CAUSES , IMPACTS AND MANAGEMENT STRATEGIES

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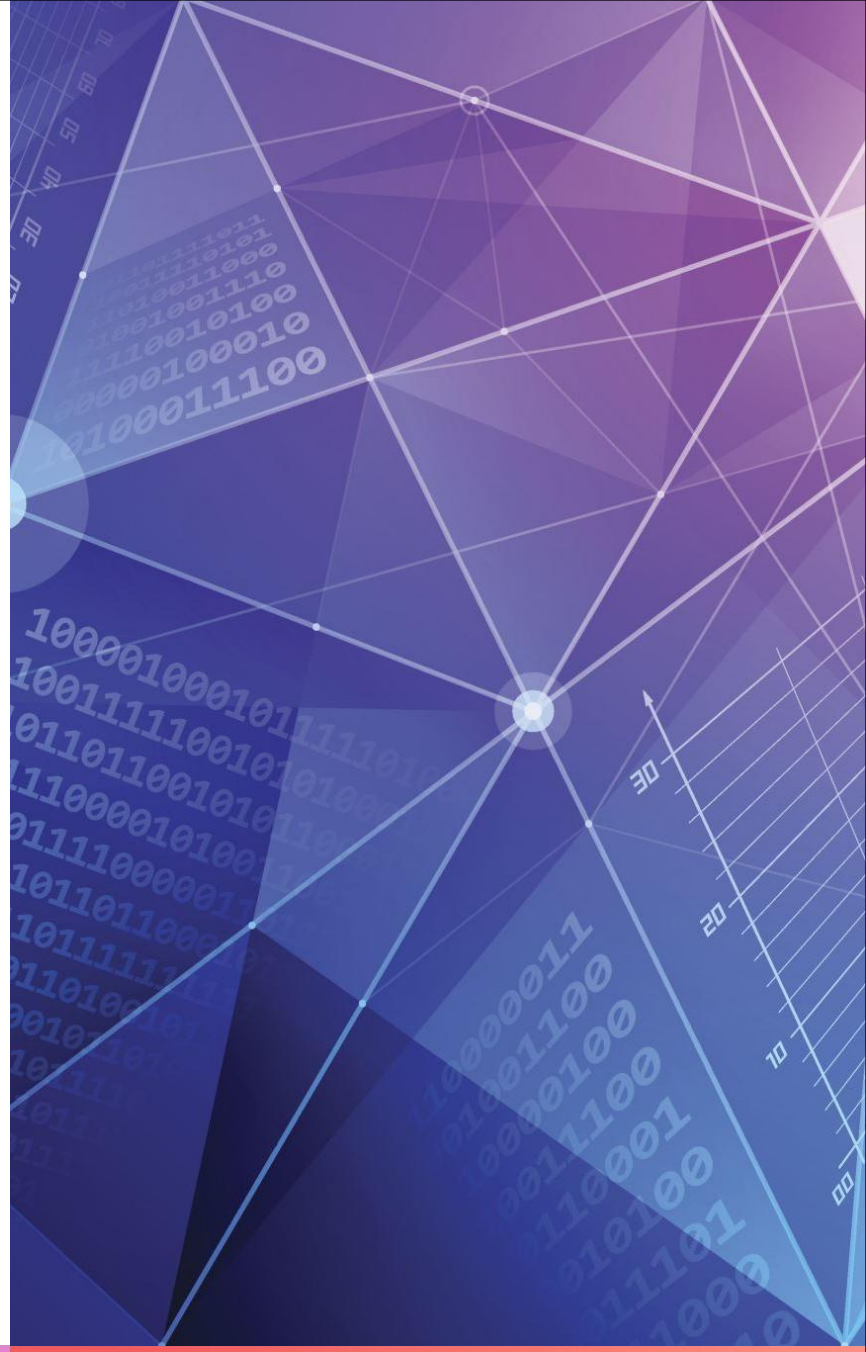
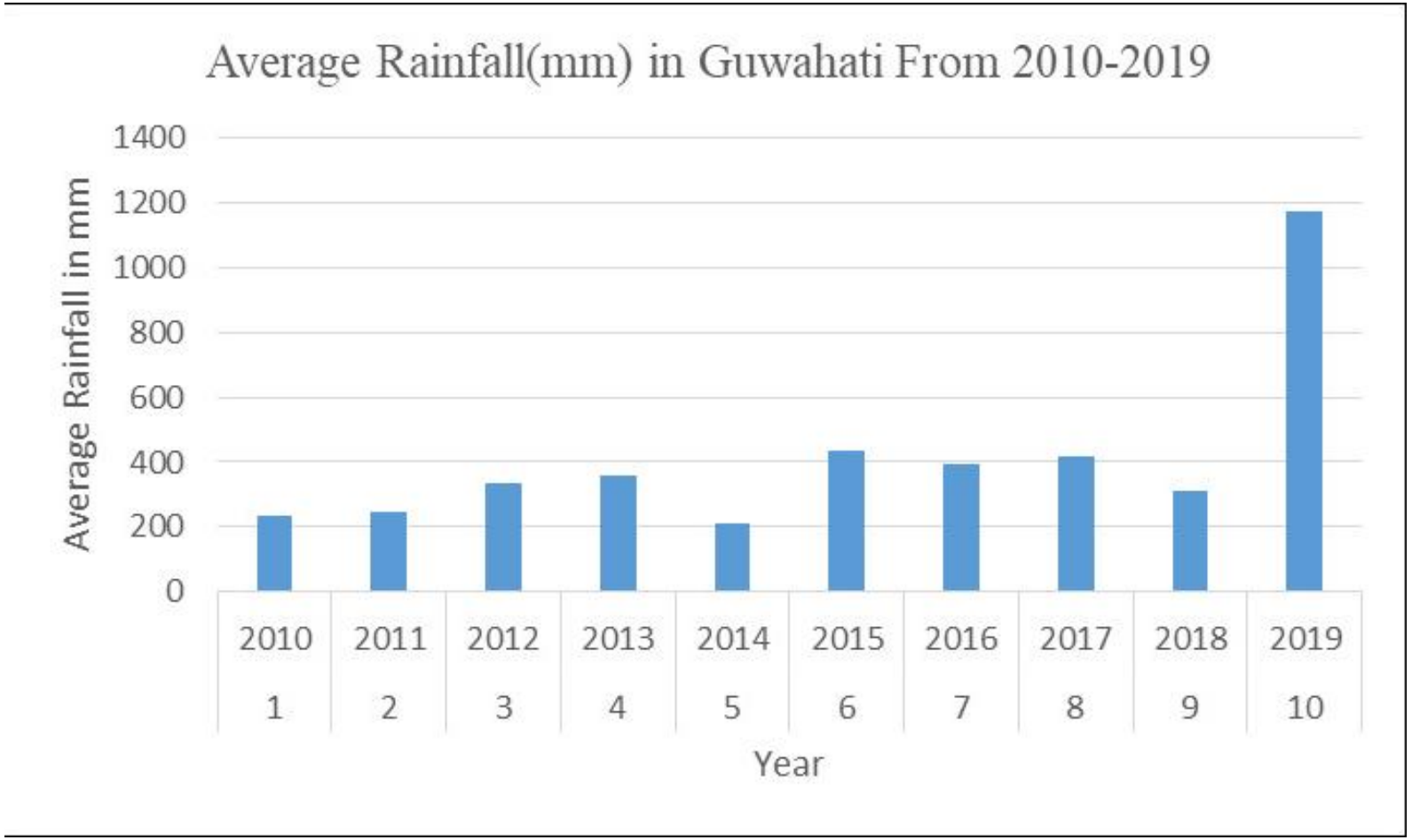


INTRODUCTION

- Guwahati faces floods almost every year.
- The problem became worse in 2024–2025 with record rainfall.
- Floods disturb daily life, damage roads and houses, affect economy and health.
- In this presentation, we will share what we learned about the causes, impacts, and management of floods.

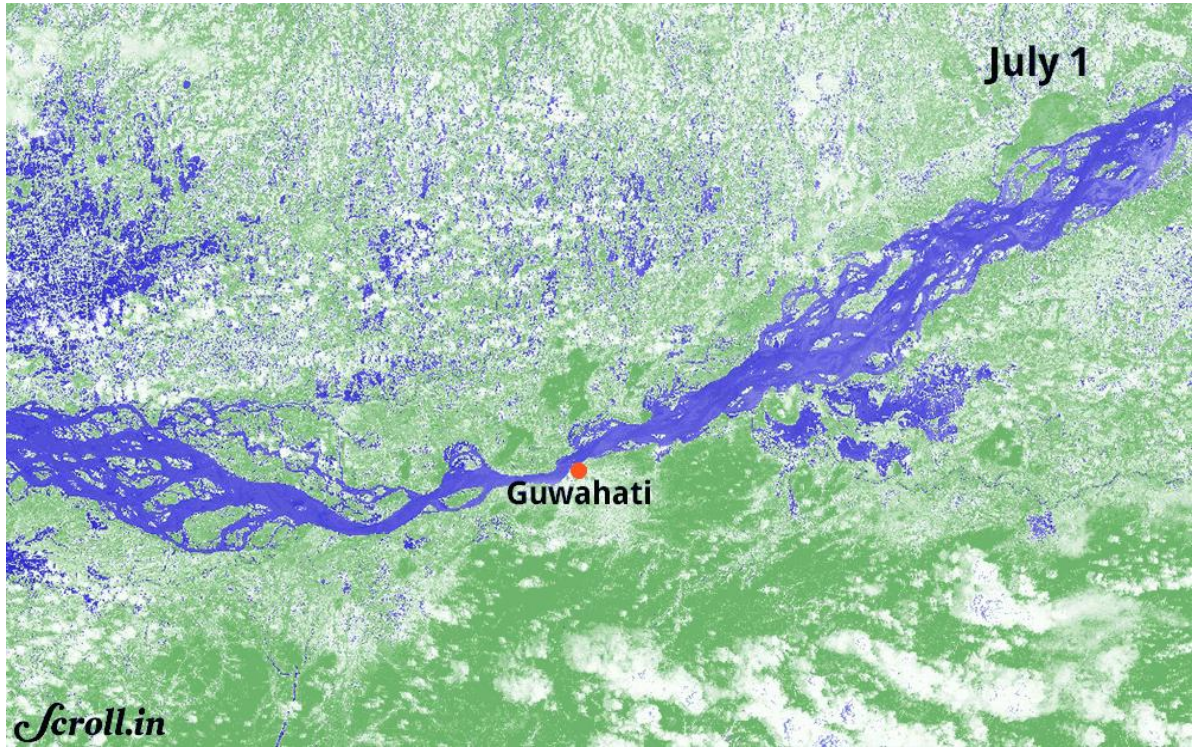
BACKGROUND: GUWAHATI'S FLOOD REALITY

- Guwahati is a bowl-shaped city, surrounded by hills, and drains into the Brahmaputra.
- May–June 2025: record rainfall (111 mm in 24 hours).
- Floods hit 18 municipal wards, affecting transport and electricity.
- Over 5 lakh people affected across Assam.



CAUSES: NATURAL FACTORS

- **Heavy Rainfall:** Sudden and very high rainfall during monsoon.
- **Runoff from Hills:** Water from Meghalaya hills flows into the city.
- **Rivers/Streams:** Bharalu, Bahini, and others overflow, especially when outlets are blocked.



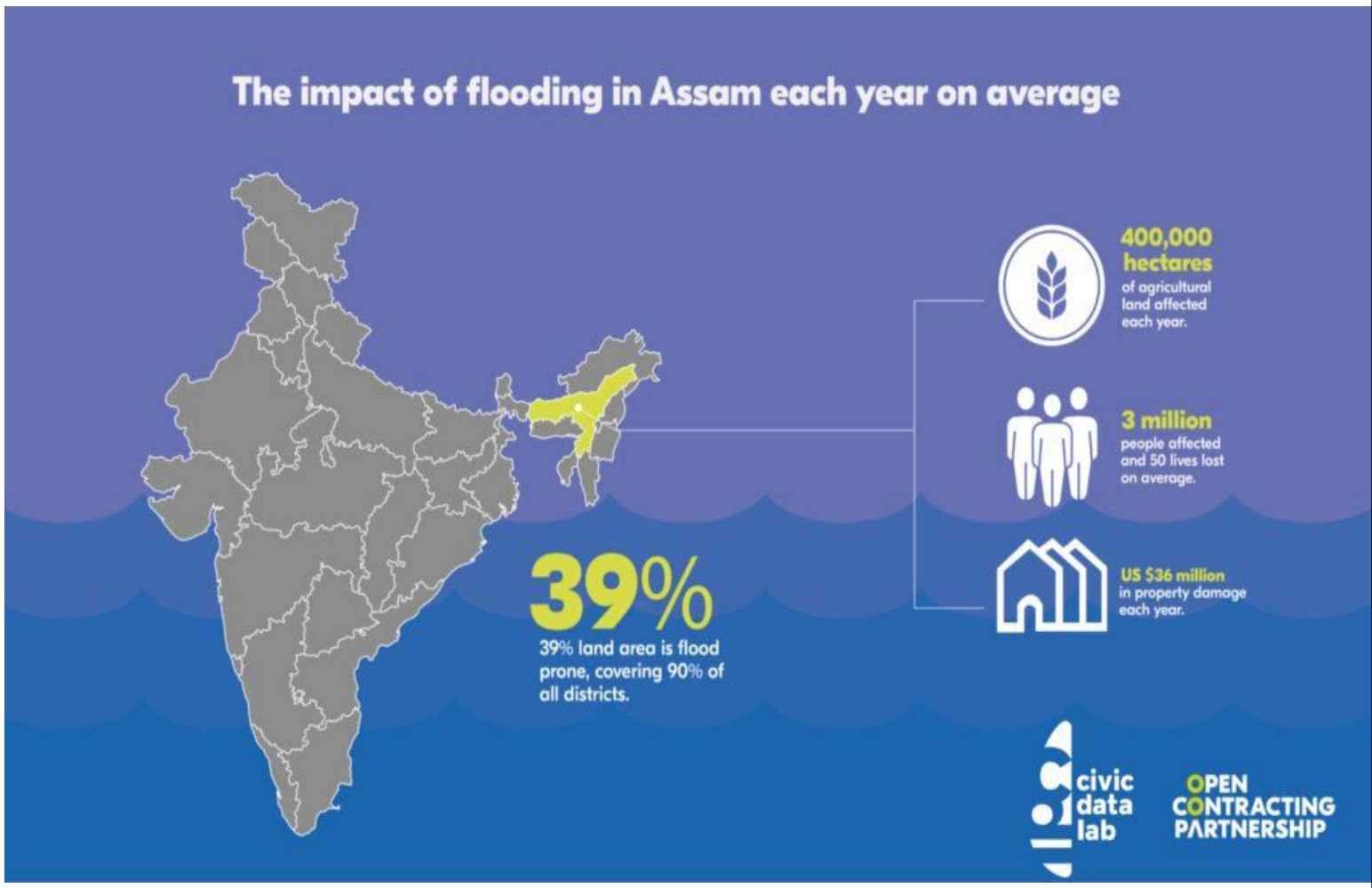
CAUSES: HUMAN FACTORS

- **Unplanned Urbanization:** Wetlands like Silsako and Deepor Beel filled for construction.
- **Poor Drainage:** Drains are small, clogged with plastic and waste.
- **Construction Issues:** Flyovers built without proper drainage results water entering homes.
- **Old Infrastructure:** Many drains and culverts are outdated and weak.

Dredging in Progress in Silsaku Beel and Pamohi Channel

CAUSES: CLIMATE CHANGE

- Rainfall patterns becoming irregular and heavier.
- Monsoon starting earlier (sometimes in May).
- Flood periods becoming longer (e.g., 7 months in 2022).
- Assam is one of the most flood-prone regions globally.



IMPACTS: INFRASTRUCTURE & DAILY LIFE

Infrastructure Damage: Severely damaged roads, collapsed sections, overflowing homes and businesses.

Transportation Paralysis: Traffic jams, stranded vehicles, suspended ferry/rail services.

Specific Example: NH-27 near Boragaon flyover, NH-6 near Jorabat severely waterlogged.

Ride-hailing services often refuse trips; public transport collapses.

Power Outages: Frequent and prolonged electricity cuts.

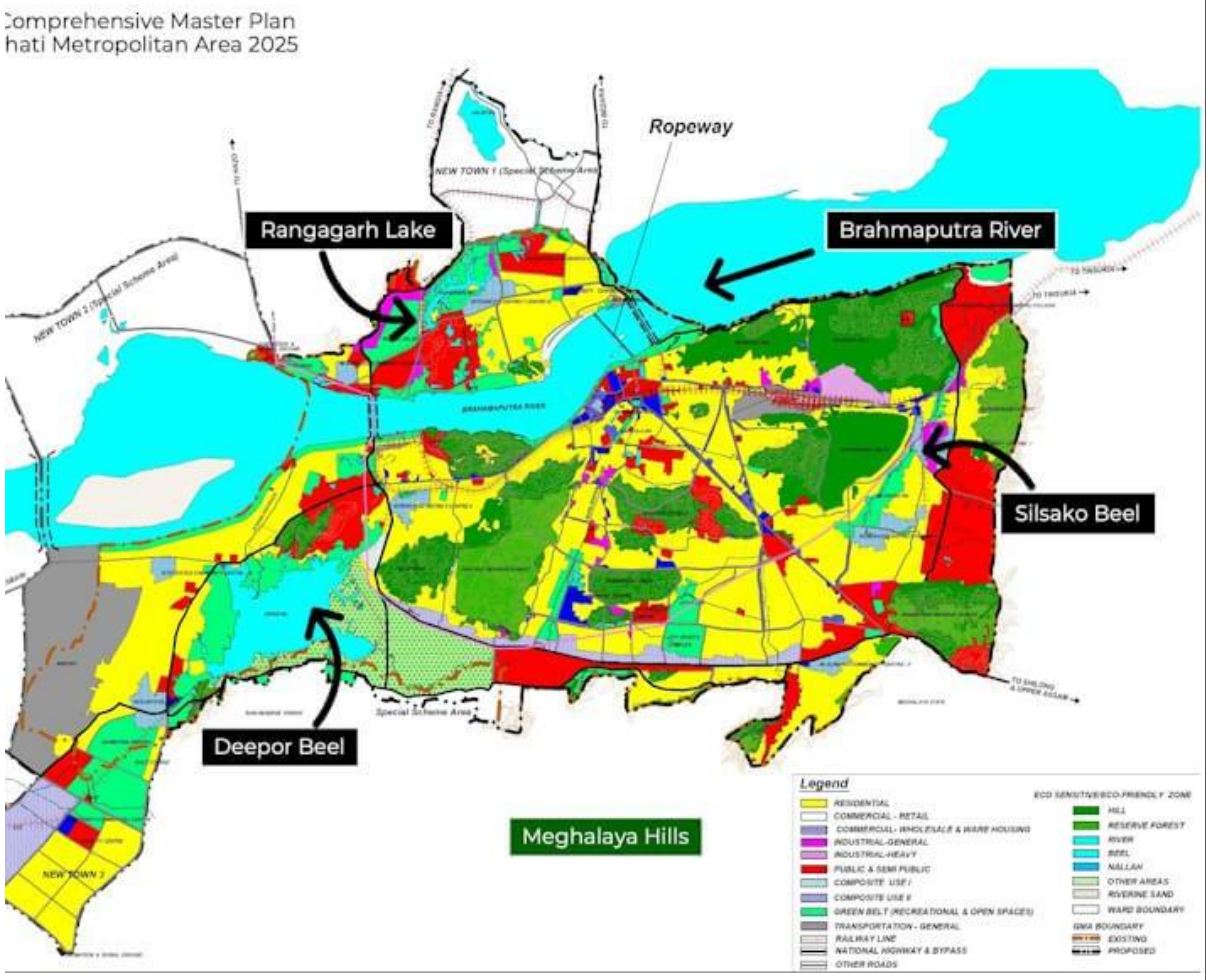
Disruption of Daily Life: Schools and offices shut and essential services affected.

IMPACTS: HEALTH & ECONOMY

- Rise in diseases like dengue, malaria, and waterborne infections.
- Small businesses and shops suffer huge losses.
- Stress and anxiety among people due to recurring floods.
- Many families displaced and forced to move to relief camps.
- Death toll from floods and landslides across Assam at 11 as of early June 2025.

CURRENT MANAGEMENT STRATEGIES

- Mission Flood-Free Guwahati started by government.
- Drains and rivers cleaned (desiltation, super sucker machines).
- High-capacity pumps installed in some areas.
- Wetland restoration (Silsako Beel, eviction of encroachments).
- New master drainage plan being drafted. by GMDA for a new city-wide system.



CHALLENGES IN MANAGEMENT

- Floods still happen even after spending money on projects.
- Drains get clogged again after every rainfall.
- Encroachment on wetlands continues.
- Lack of coordination between departments.
- Old infrastructure not suitable for today’s rainfall.

RECOMMENDATIONS: PLANNING & RESTORATION

Master Drainage Plan: The city needs a single, proper drainage plan that covers all areas. Right now, only small local solutions are made, which do not solve the problem fully.

Wetland Protection: Wetlands like Silsako Beel and Deepor Beel should be restored and protected. These natural water bodies can store extra rainwater. Encroachment on wetlands should be strictly stopped.

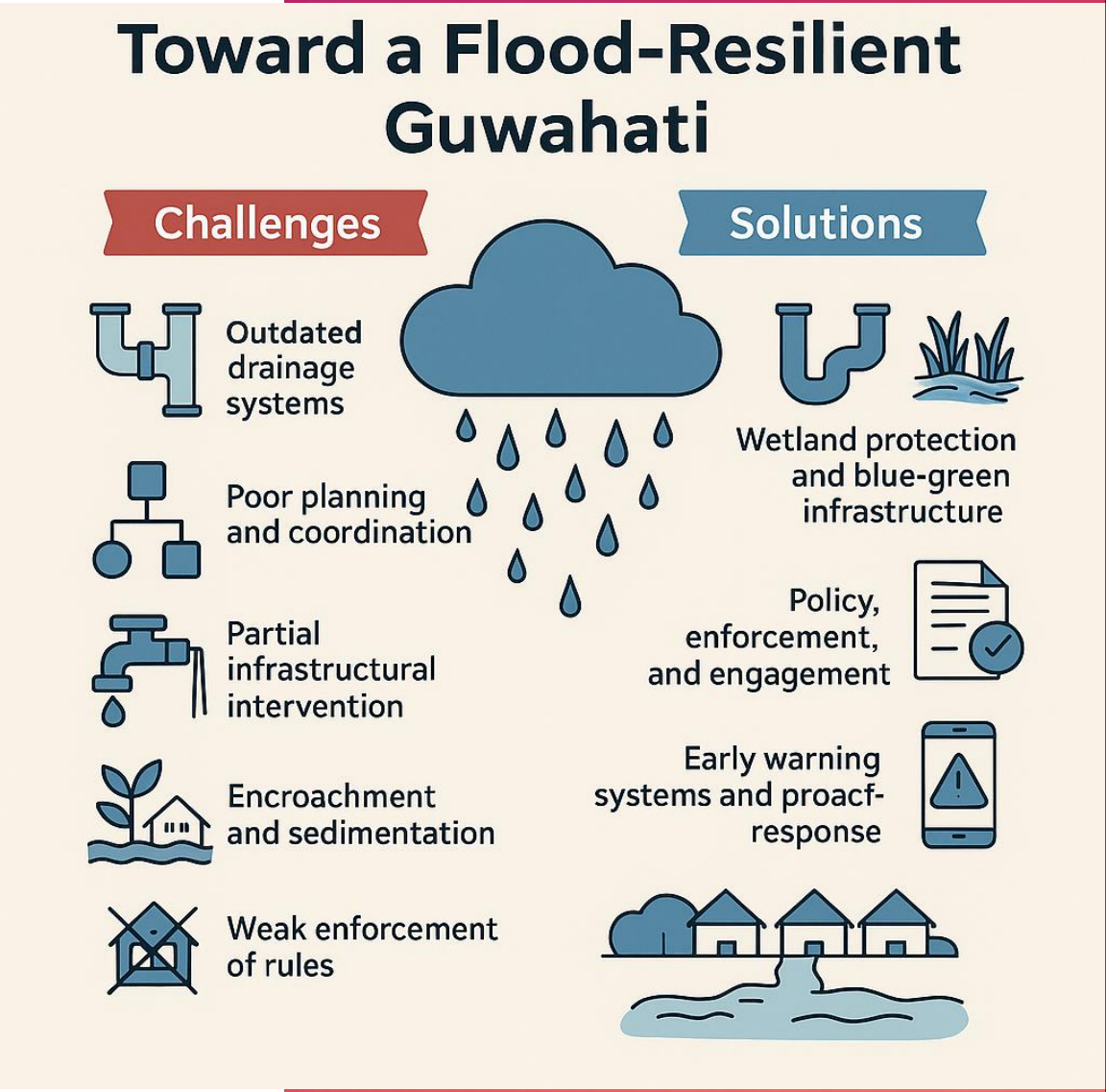
Drainage Maintenance: Drains should be cleaned regularly, not just before the rainy season. Machines can be used for faster desiltation, and cleaning should happen throughout the year.

RECOMMENDATIONS: ENFORCEMENT& ENGAGEMENT

- **Waste Management:** Plastic and other waste should be reduced and recycled. This will stop drains from getting clogged and reduce waterlogging.
- **Climate-Friendly Planning:** When making new roads or buildings, future climate and rainfall should be considered. Green ideas like pavements that absorb water and small rain gardens can also be used.
- **Strict Rules and Responsibility:** Rules should be followed properly, and those who break them should be punished. Authorities must also take full responsibility for flood control.
- **Public Awareness:** People should be made more aware about keeping drains clean, not throwing garbage, and protecting wetlands. Citizens should also help in reporting and monitoring.
- **Use of Technology:** Modern tools like maps, sensors, and warning systems can help predict floods and give alerts in time.

CONCLUSION

- Flooding in Guwahati is caused by both natural reasons and human activities.
- The problem is getting worse due to rapid growth and climate change.
- From our internship, we learned that solving floods needs planning, public awareness, and cooperation between government and people.
- Only through joint effort can Guwahati become safer and flood-resilient.



THANK YOU